

Sinhorn Div. Interpolates between OT and MMD

Interpolation in terms of:

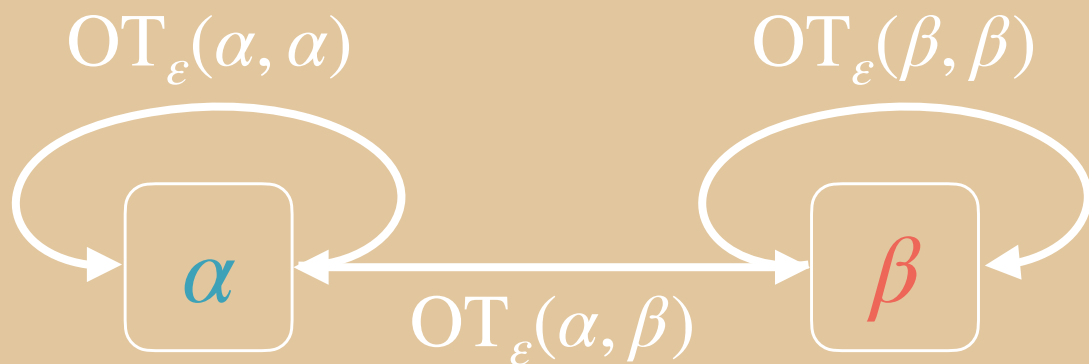
- Couplings (sparse vs flat/uniform)
- Computational complexity ($\mathcal{O}(n^3 \log n)$ vs $\mathcal{O}(n^2)$)
- Statistical complexity ($\mathcal{O}(n^{-\frac{1}{d}})$ vs $\mathcal{O}(n^{-\frac{1}{2}})$)

Wasserstein Barycenters

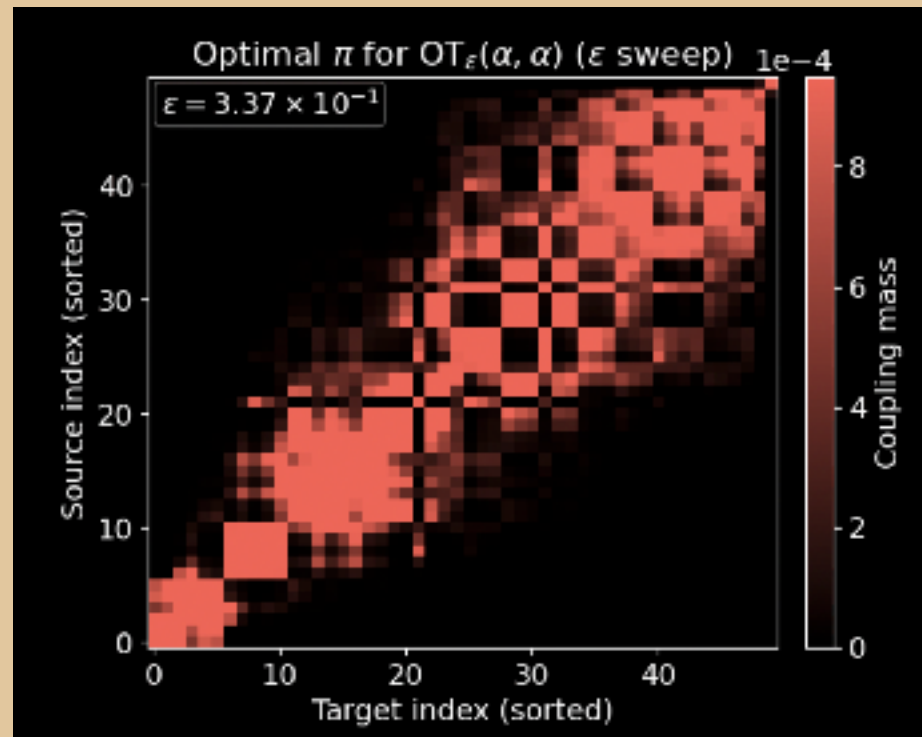
- Extends means/centroids to distributions
- Like Frechet mean, but for distributions
- Barycenter: $\nu^\star = \arg \min_{\nu} \sum_{i=1}^K \lambda_i W_p^p(\mu_i, \nu)$

Sinkhorn Divergence: Motivation

- $W_\varepsilon(\alpha, \alpha) > 0$
- Non-diagonal coupling
- Subtract self-costs



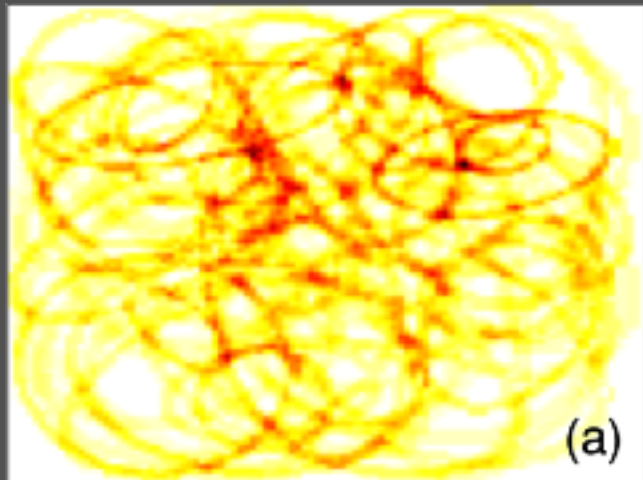
- Yields sinkhorn divergence



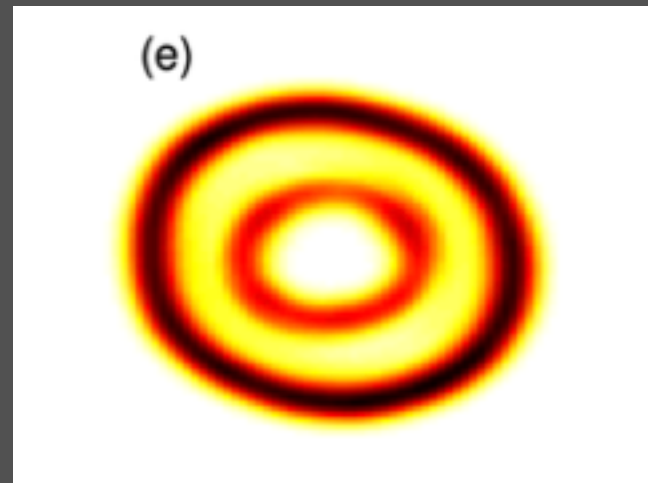
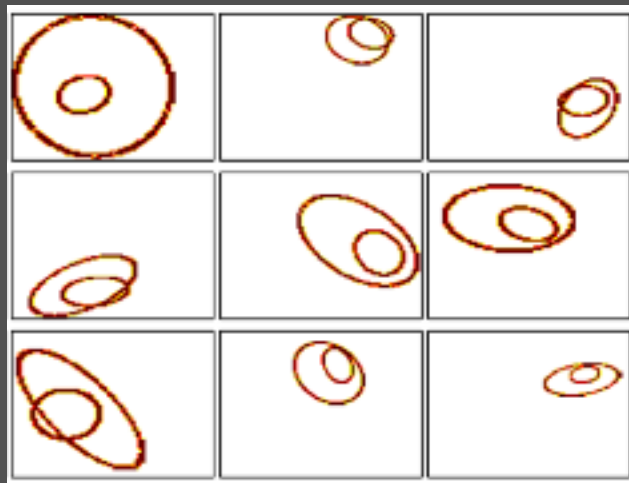
Sinhorn Divergence:

$$S_{\varepsilon}(\alpha, \beta) = OT_{\varepsilon}(\alpha, \beta) - \frac{1}{2}OT_{\varepsilon}(\alpha, \alpha) - \frac{1}{2}OT_{\varepsilon}(\beta, \beta)$$

OT baricenters consistent with geometry



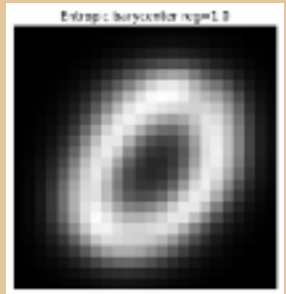
Euclidean averaging



OT Averaging

Barycenter Computation

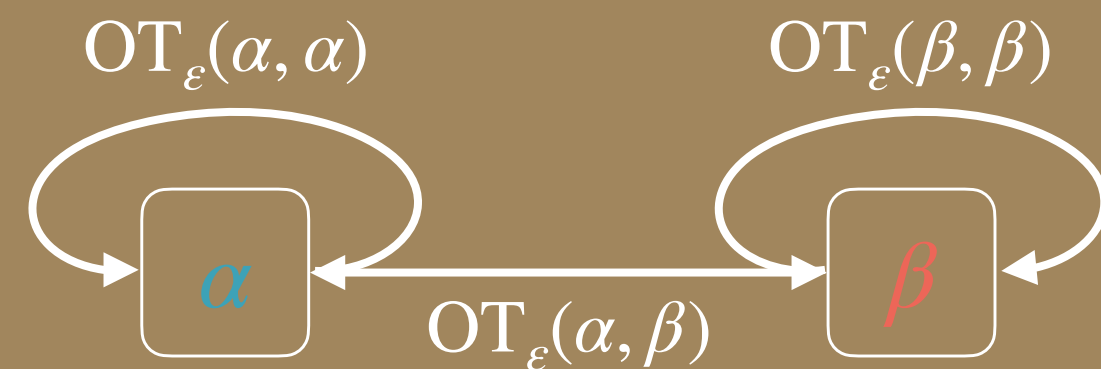
- Exact form is (very large) LP - expensive
- Add entropy regularization, use Sinkhorn
- Entropic barycenters: “smooth”, scalable



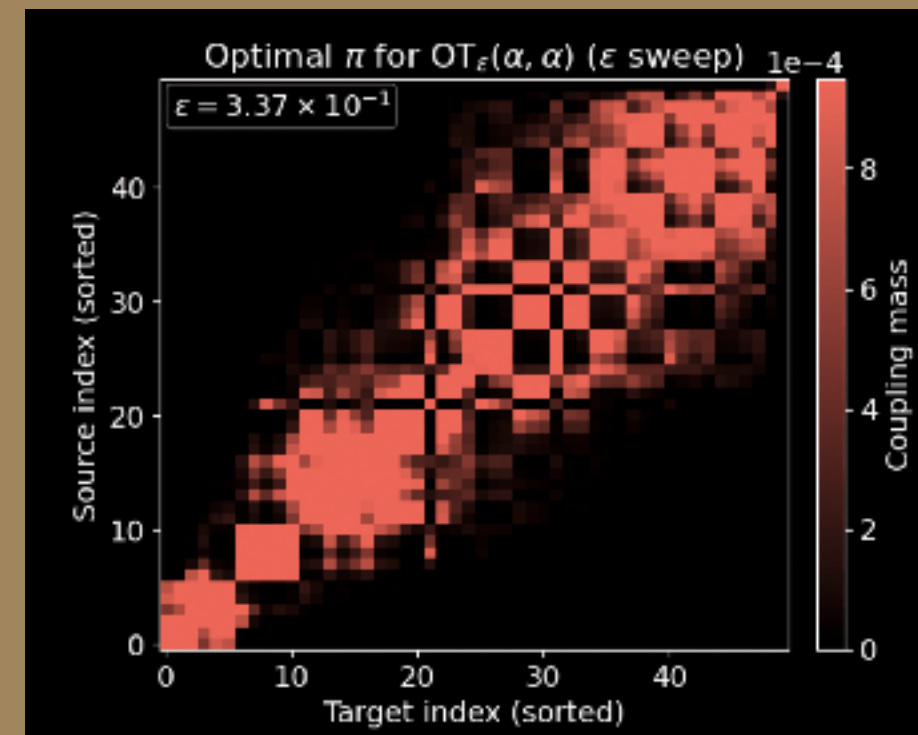
Last Time

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Sinhorn Divergence:

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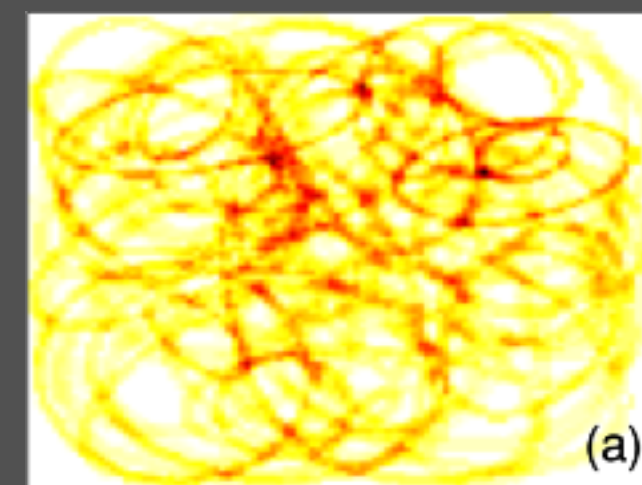
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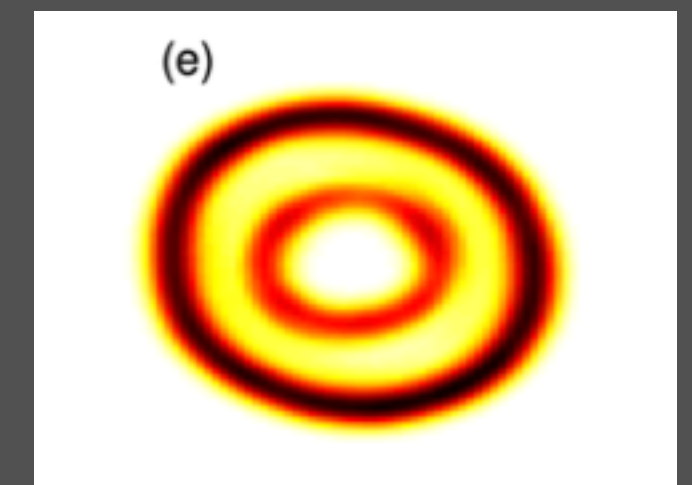
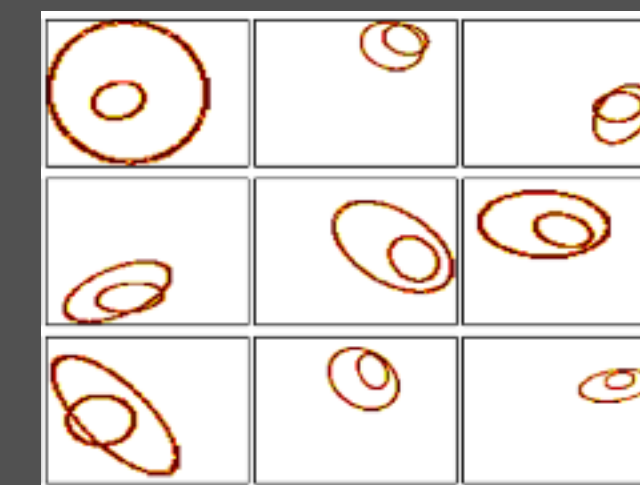
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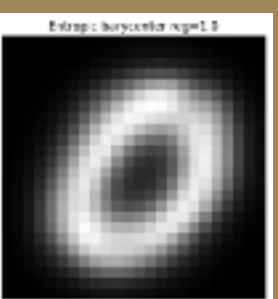
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Today